

5634082

B.Tech. DEGREE EXAMINATION, MAY 2018.

Fourth Semester

Computer Science and Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

(2009 – 2012 and 2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. List out the criteria in algorithms.
2. Prove if $f(n) = a_m n^m + \dots + a_1 n + a_0$, then $f(n) = O(n^m)$.
3. Write the algorithm for merge sort.
4. Write a control abstraction for the ordering paradigm.
5. Note down the function to compute lengths of shortest paths.
6. Differentiate AND graph with OR graph.

7. What are the mechanisms for backtracking algorithm?
8. What is Hamiltonian cycle?
9. What is least cost search algorithm?
10. Write the difference between NP-hard and NP-completeness problem?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Examine the performance of recursive algorithm with an example. (11)

Or

12. Analysis the time and space complexity of binary search algorithm. (11)

UNIT II

13. Discuss about divide and conquer technique using the algorithm of maximum and minimum finding. (11)

Or

14. Express about greedy method using minimum spanning tree. (11)

UNIT III

15. Discuss briefly about multistage graph pseudo code corresponding to forward and backward approach. (11)

Or

16. Explain the algorithm for 0/1 knapsack problem. (11)

UNIT IV

17. What is 8-queens problem? Discuss it using backtracking algorithm. (11)

Or

18. Analysis in detail about graph coloring. (11)

UNIT V

19. Write the algorithm for travelling salesman problem and test the correctness of the algorithm using suitable test data. (11)

Or

20. Describe briefly about NP-Hard problem with suitable algorithm. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2016.

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(2009-2012 Batches)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is an algorithm? Give an example.
2. What is time complexity? Give an example.
3. What is divide and conquer method?
4. What is binary search?
5. Define an adjacency matrix.
6. How does a graph differ from a tree?
7. What is graph coloring?
8. What is back tracking? Where it is used?

9. Define NP-hard problem.
10. What is NP-complete problem?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, One from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the various mathematical notations used in program in detail.

Or

12. Discuss about selection sort algorithm with an example.

UNIT II

13. Write an algorithm to find the maximum and minimum number of given n numbers.

Or

14. Write and explain the minimum cost spanning tree algorithm.

UNIT III

15. What is dynamic programming? Discuss an algorithm with an example.

Or

16. What are AND/OR graphs? Explain in detail.

UNIT IV

17. Write an algorithm to solve the 8 queen problem by using back tracking method.

Or

18. Discuss about Hamiltonian cycle in detail.

UNIT V

19. Explain about travelling salesman problem in detail.

Or

20. Discuss one Branch and Bound method with an example.

16. (a) Define bi-connected components. Explain the impact of it in a graph. (5)
 (b) What is an articulation point? Explain with an example the effect of removing an articulation point from a graph. (6)

UNIT IV

17. Solve the sum of subset problem using backtracking techniques. What are the issues associated with the backtracking techniques? (11)

Or

18. Draw a Hamiltonian circuit. Describe its meaning. How does it play a role in the real world application? (11)

UNIT V

19. (a) Define branch and bound technique. What will be the criteria for the branching and what will be the criteria for the bounding? (6)
 (b) Outline the procedure for solving the travelling salesman problem using branch and bound technique. (5)

Or

20. (a) Explain how the 0/1 knapsack problem is solved using branch and bound technique. (6)
 (b) Write down the control abstraction for the Least Cost search. (5)

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DESIGN AND ANALYSIS OF ALGORITHMS

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Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define time complexity of an algorithm? What are the notations used for this?
2. Find the order (big O) of an algorithm having $T(n) = n^4 + 7n + 3$, where 'n' is the size of input.
3. Enumerate the advantages of divide and conquer technique.
4. Assume that you have 7 containers whose weights are 40, 50, 10, 30, 20, 60 and 5 and a ship whose capacity is 120. Use the greedy technique to find an optimal solution to the container loading problem.

5. Compare greedy and dynamic programming algorithm.
6. What is the time complexity of searching an element in a binary search tree with n number of elements?
7. State 8-queen's problem.
8. What are the key issues in graph coloring problem?
9. Compare branch and bound LC and FIFO methods.
10. Define NP-hard problem.

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Write an algorithm for linear search and compare its efficiency for best case, average case and worst case inputs. (6)
- (b) Write down the heap sort algorithm and analyze its time complexity. (5)

Or

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12. (a) Devise a non recursive algorithm for finding minimum and maximum in a list of N elements and analysis its time complexity. (6)
- (b) Outline the flow of the execution of binary search for the given input of size $n = 10$ with the following elements 7, 12, 45, 67, 78, 89, 95, 120, 156, 430 where the key element to be searched is 89. (5)

UNIT II

13. Design a control abstraction for general divide and conquer technique and explain with an example how this algorithm reduces its complexity. (11)

Or

14. State knapsack problem. Write down the greedy algorithm for solving a knapsack problem and explain with an example. (11)

UNIT III

15. (a) What is a multistage graph? How do you deploy the multistage graph to solve all shortest path problem? (6)
- (b) State travelling salesman problem. How do you use dynamic programming technique to solve this problem? (5)

Or

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2016.

Fourth Semester

Computer Science and Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Define small oh notation.
2. What is meant by worst case analysis? Give example.
3. Define Masters 'Theorem.
4. What is a Greedy Technique?
5. Give the advantage of Dynamic Programming.
6. What is a Topological sort?
7. What is a Hamiltonian Cycle?
8. Define Back Tracking.
9. What is a Least Cost search?
10. What is meant by NP Hard problems?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. Give the Insertion Sort Algorithm. Perform the Analysis of the Algorithm.

Or

12. Give the Steps for the Analysis of recursive Algorithms. Finding the formula for finding the Fibonacci Series.

UNIT II

13. Explain binary search algorithm using Divide and Conquer method and discuss about the Worst Case, Best Case and Average Case.

Or

14. Discuss the Prim's method for Finding the Minimum Spanning Tree.

UNIT III

15. Discuss Dijkstra's all pair shortest algorithm.

Or

16. Explain the chained Matrix multiplication using Dynamic programming technique.

UNIT IV

17. Solve the 8 Queens problem using back tracking.

Or

18. Explain sum of subsets problem and give the algorithm to find the same.

UNIT V

19. Explain the Travelling Salesman Problem and solve for five cities using Branch and Bound.

Or

20. Discuss in detail about NP- Hard and NP- Completeness.

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DESIGN AND ANALYSIS OF ALGORITHMS

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Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. What is theta notation?
2. Define recurrence relation.
3. What is a knapsack problem?
4. List some choices for the selection of Pivot element in Quick Sort.
5. What are biconnected components?
6. Define topological ordering.

7. What is meant by backtracking?
8. What is a Hamiltonian cycle?
9. What is NP Hard Problem?
10. What is meant by Bounding?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

All questions carry equal marks.

UNIT I

11. Explain the Heap sort algorithm and give the best, worst and average case analysis.

Or

12. Derive the formula for Fibonacci Series.

UNIT II

13. Explain Strassen's matrix multiplication with an example.

Or

14. Explain Kruskal's minimum spanning tree algorithm.

UNIT III

15. Explain all pairs shortest path algorithm using Dynamic Programming.

Or

16. Explain the Graph Traversal Techniques.

UNIT IV

17. Explain Backtracking and how the 8 queens problem is solved using the same.

Or

18. Discuss the applications of graph coloring and how it is solved.

UNIT V

19. Explain the LC search with an example and the control abstractions for the same.

Or

20. Explain the solution to 0/1 Knapsack problem.